Monitor performance

Cloud Manager 3.8

NetApp
May 31, 2021

This PDF was generated from https://docs.netapp.com/us-en/occm38/concept_monitoring.html on May 31, 2021. Always check docs.netapp.com for the latest.

Table of Contents

Monitor performance . . . 1
Learn about the Monitoring service . . . 1
Monitoring Cloud Volumes ONTAP in AWS . . . 3

Monitor performance

Learn about the Monitoring service

By leveraging the NetApp Cloud Insights service, Cloud Manager gives you insights into the health and performance of your Cloud Volumes ONTAP instances and helps you troubleshoot and optimize the performance of your cloud storage environment.

Features

- Automatically monitor all volumes
- View volume performance data in terms of IOPS, throughput, and latency
- Identify performance issues to minimize impact on your users and apps

Supported cloud providers

The Monitoring service is supported with Cloud Volumes ONTAP for AWS.

Cost

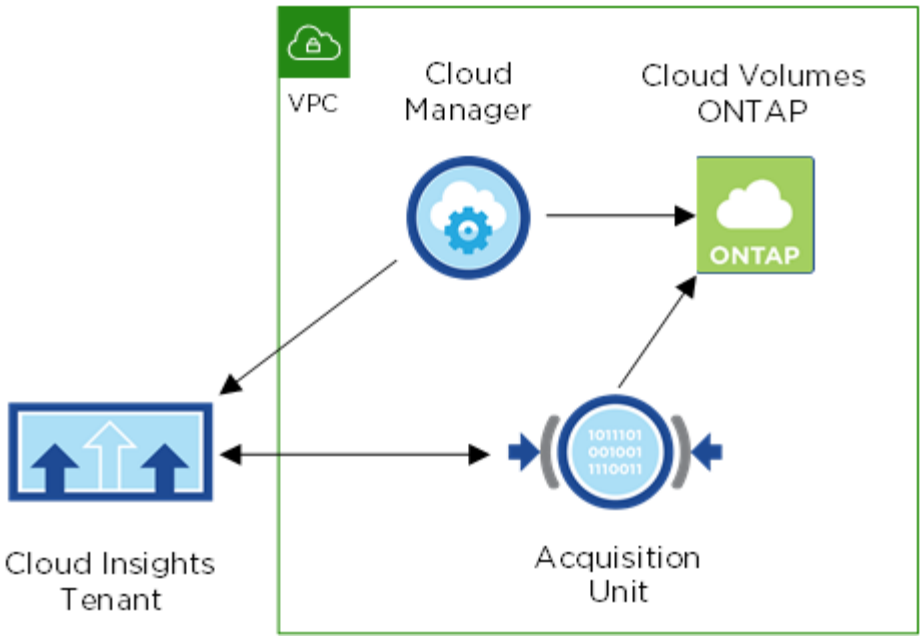
Monitoring is currently available as a Preview. Activation is free, but Cloud Manager launches a virtual machine in your VPC to facilitate monitoring. This VM results in charges from your cloud provider.

How Cloud Insights works with Cloud Manager

At a high-level, Cloud Insights integration with Cloud Manager works like this:

1. You enable the Monitoring service on Cloud Volumes ONTAP.
2. Cloud Manager configures your environment. It does the following:
    1. Creates a Cloud Insights tenant (also called *environment*) and associates all users in your Cloud Central account to the tenant.
    2. Enables a 30-day free trial of Cloud Insights.
    3. Deploys a virtual machine in your VPC called an Acquisition Unit, which facilitates monitoring of volumes (this is the VM mentioned in the Cost section above).
    4. Connects the Acquisition Unit to Cloud Volumes ONTAP and to the Cloud Insights tenant.
3. In Cloud Manager, you click Monitoring and use the performance data to troubleshoot and optimize performance.

The following image shows the relationship between these components:



The Acquisition Unit

When you enable Monitoring, Cloud Manager deploys an Acquisition Unit in the same subnet as the Connector.

An *Acquisition Unit* collects performance data from Cloud Volumes ONTAP and sends it to the Cloud Insights tenant. Cloud Manager then queries that data and presents it to you.

Note the following about the Acquisition Unit instance:

- The Acquisition Unit runs on a t3.xlarge instance with a 100 GB GP2 volume.
- The instance is named *AcquisitionUnit* with a generated hash (UUID) concatenated to it. For example: *AcquisitionUnit-FAN7FqeH*
- Only one Acquisition Unit is deployed per Connector.
- The instance must be running to access performance information in the Monitoring tab.

Cloud Insights tenant

Cloud Manager sets up a *tenant* for you when you enable Monitoring. A Cloud Insights tenant enables you to access the performance data that the Acquisition Unit collects. The tenant is a secure data partition within the NetApp Cloud Insights service.

Cloud Insights web interface

The Monitoring tab in Cloud Manager provides basic performance data for your volumes. You can go to the Cloud Insights web interface from your browser to perform more in-depth monitoring and to configure alerts for your Cloud Volumes ONTAP systems.

Free trial and subscription

Cloud Manager enables a 30-day free trial of Cloud Insights to provide performance data within Cloud Manager and for you to explore the features that Cloud Insights Standard Edition has to offer.

You need to subscribe by the end of the free trial or your Cloud Insights tenant will eventually be deleted. You

can subscribe to either the Basic, Standard, or Premium edition to continue using the Monitoring feature within Cloud Manager.

Learn how to subscribe to Cloud Insights.

Monitoring Cloud Volumes ONTAP in AWS

Complete a few steps to get started monitoring Cloud Volumes ONTAP performance.

Quick start

Get started quickly by following these steps or scroll down to the remaining sections for full details.

1 Verify support for your configuration

You need a new installation of Cloud Manager 3.8.4 or later in AWS, Cloud Volumes ONTAP in AWS, and you must be a new Cloud Insights customer.

2 Enable Monitoring on your new or existing system

- New working environments: Be sure to keep Monitoring enabled when you create the working environment (it's enabled by default).
- Existing working environments: Select a working environment and click **Start Monitoring**.

3 View performance data

Click **Monitoring** and view performance data for your volumes.

4 Subscribe to Cloud Insights

Subscribe before your 30-day free trial ends to continue seeing performance data within Cloud Manager and Cloud Insights. Learn how to subscribe.

Requirements

Read the following requirements to make sure that you have a supported configuration.

Supported Cloud Manager versions

You need a new installation of Cloud Manager 3.8.4 or later. A new installation is needed because a new infrastructure is required to enable the Monitoring service. This infrastructure is available starting with new installations of Cloud Manager 3.8.4.

Supported Cloud Volumes ONTAP versions

Any version of Cloud Volumes ONTAP in AWS.

Cloud Insights requirement

You must be a new Cloud Insights customer. Monitoring isn't supported if you already have a Cloud Insights tenant.

Email address for Cloud Central

The email address for your Cloud Central user account should be your business email address. Free email domains like gmail and hotmail aren't supported when creating a Cloud Insights tenant.

Networking for the Acquisition Unit

The Acquisition Unit uses 2-way/mutual authentication to connect to the Cloud Insights server. The client certificate must be passed to the Cloud Insights server to be authenticated. To accomplish this, the proxy must be set up to forward the http request to the Cloud Insights server without decrypting the data.

The Acquisition Unit uses the following two endpoints to communicate with Cloud Insights. If you have a firewall between the Acquisition Unit server and Cloud Insights, you need these endpoints when configuring firewall rules:

```
https://aulogin.<Cloud Insights Domain>
https://<your-tenant-ID>.<Cloud Insights Domain>
```

For example:

```
https://aulogin.c01.cloudinsights.netapp.com
https://cg0c586a-ee05-45rb-a5ac-
333b5ae7718d7.c01.cloudinsights.netapp.com
```

Contact us through the in-product chat if you need help identifying your Cloud Insights domain and tenant ID.

Networking for the Connector

Similar to the Acquisition Unit, the Connector must have outbound connectivity to the Cloud Insights tenant. But the endpoint that the Connector contacts is slightly different. It contacts the tenant host URL using the shortened tenant ID:

```
https://<your-short-tenant-ID>.<Cloud Insights Domain>
```

For example:

```
https://abcd12345.c01.cloudinsights.netapp.com
```

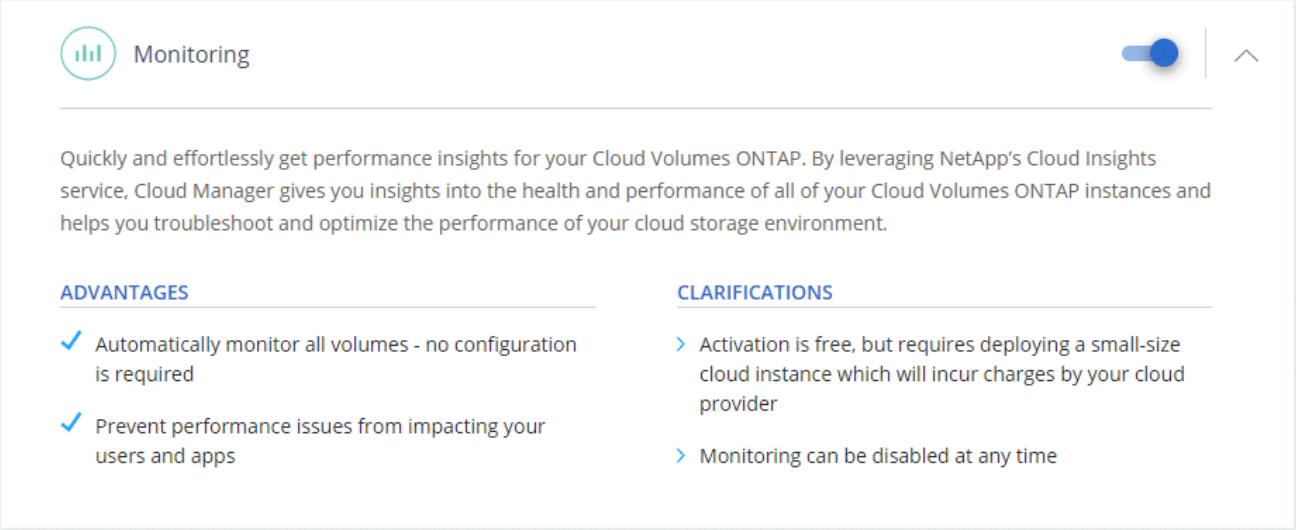
Again, you can contact us through the in-product chat if you need help identifying the tenant host URL.

Enabling monitoring on a new system

The Monitoring service is enabled by default in the working environment wizard. Be sure to keep the option enabled.

Steps

1. Click **Create Cloud Volumes ONTAP**.
2. Select Amazon Web Services as the cloud provider and then choose a single node or HA system.
3. Fill out the Details & Credentials page.
4. On the Services page, leave the service enabled and click **Continue**.

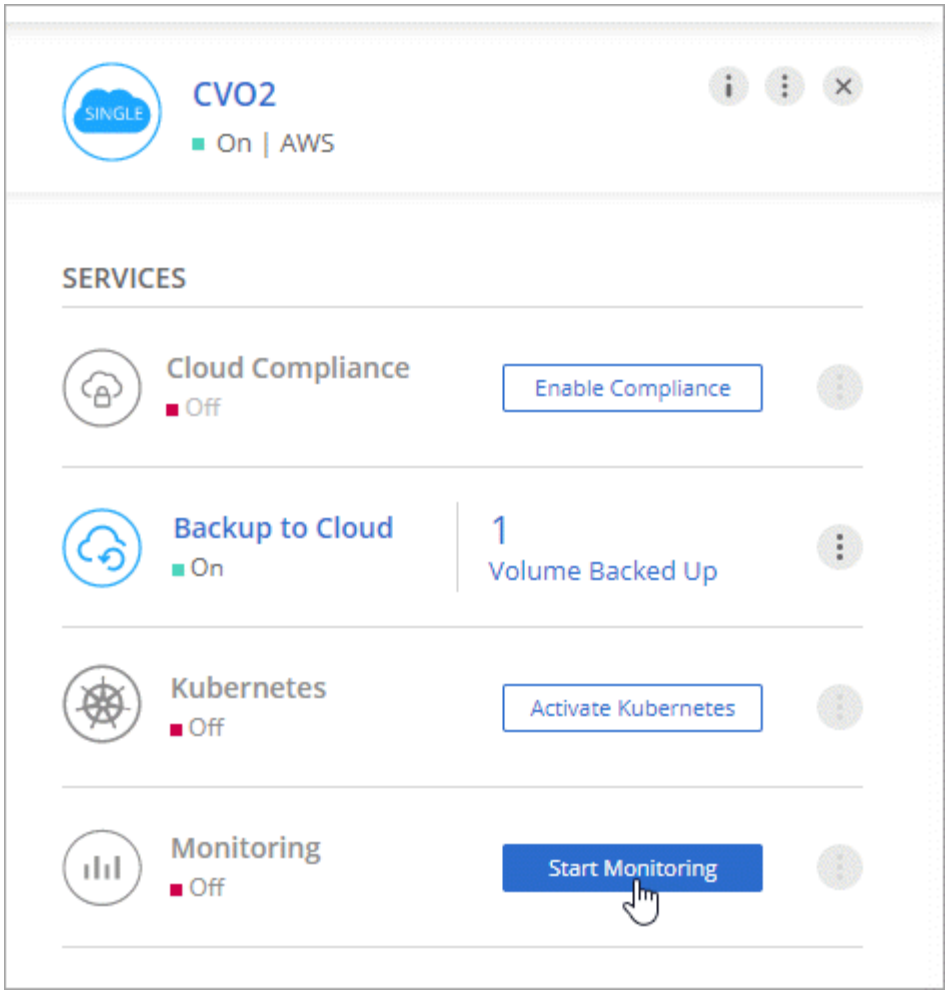


Enabling monitoring on an existing system

Enable monitoring at any time from the working environment.

Steps

1. At the top of Cloud Manager, click **Working Environments**.
2. Select a working environment.
3. In the pane on the right, click **Start Monitoring**.



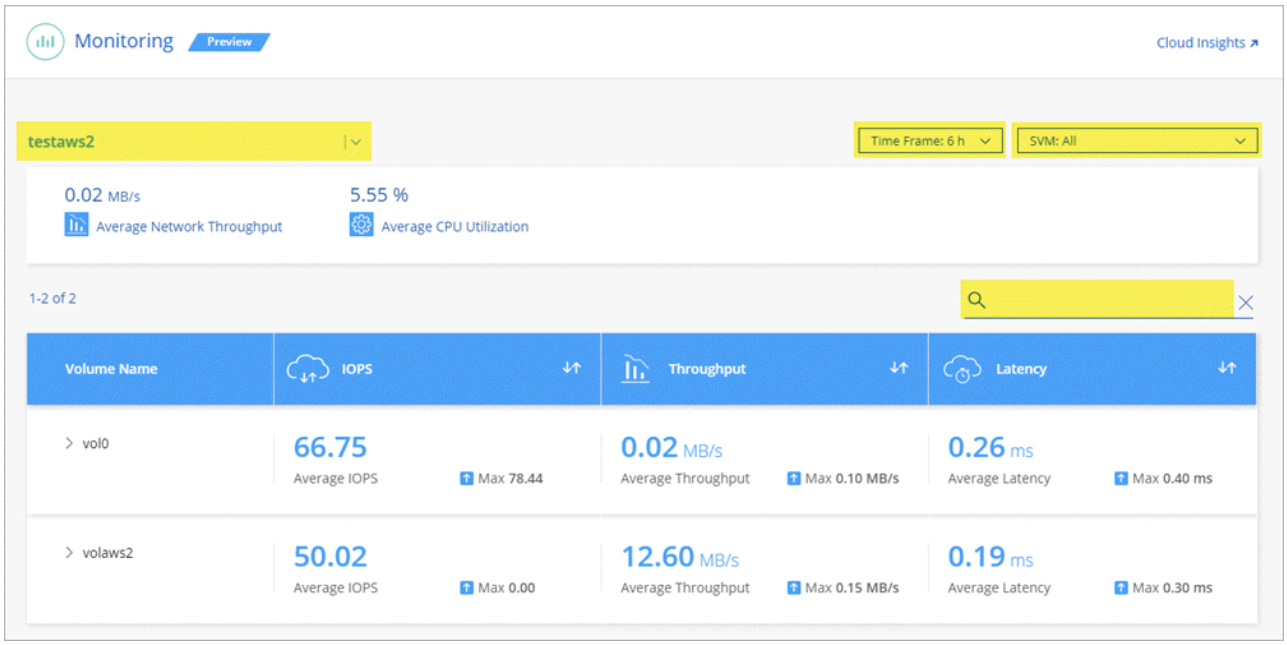
Monitoring your volumes

Monitor performance by viewing IOPS, throughput, and latency for each of your volumes.

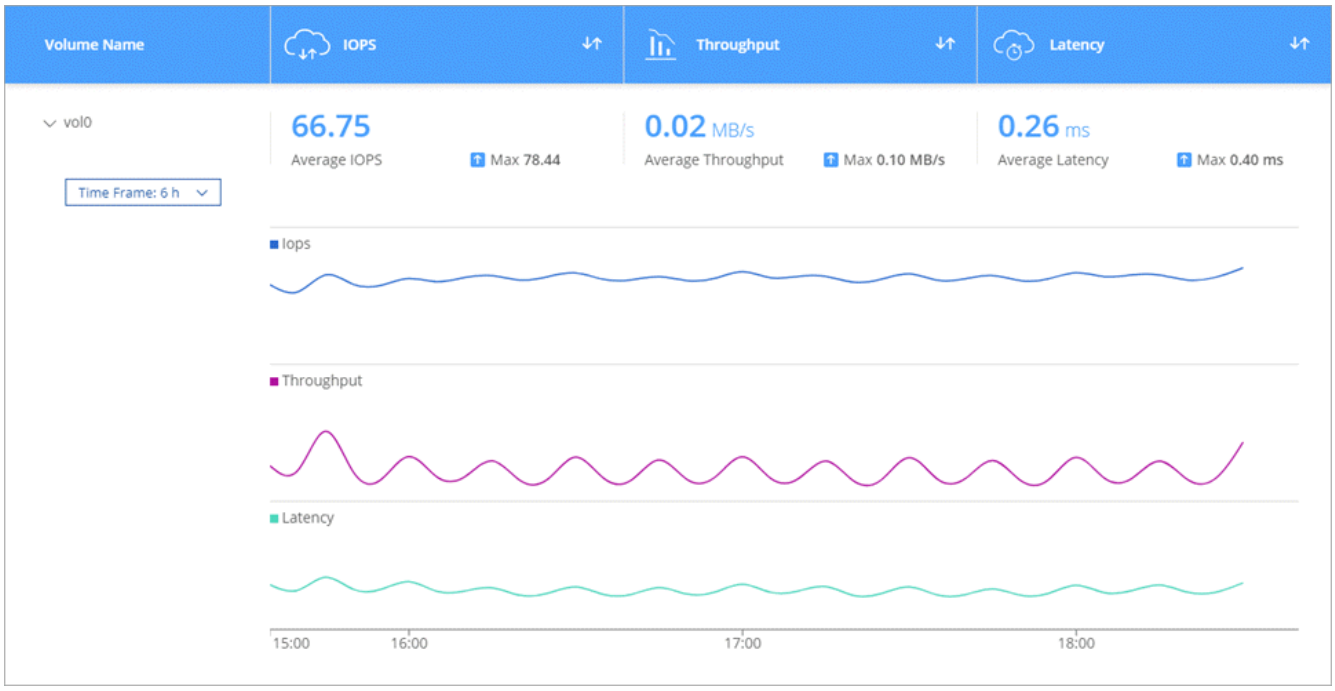
Steps

1. At the top of Cloud Manager, click **Monitoring**.
2. Filter the contents of the dashboard to get the information that you need.
   - Select a specific working environment.
   - Select a different timeframe.
   - Select a specific SVM.
   - Search for a specific volume.

     The following image highlights each of these options:



3. Click a volume in the table to expand the row and view a timeline for IOPS, throughput, and latency.



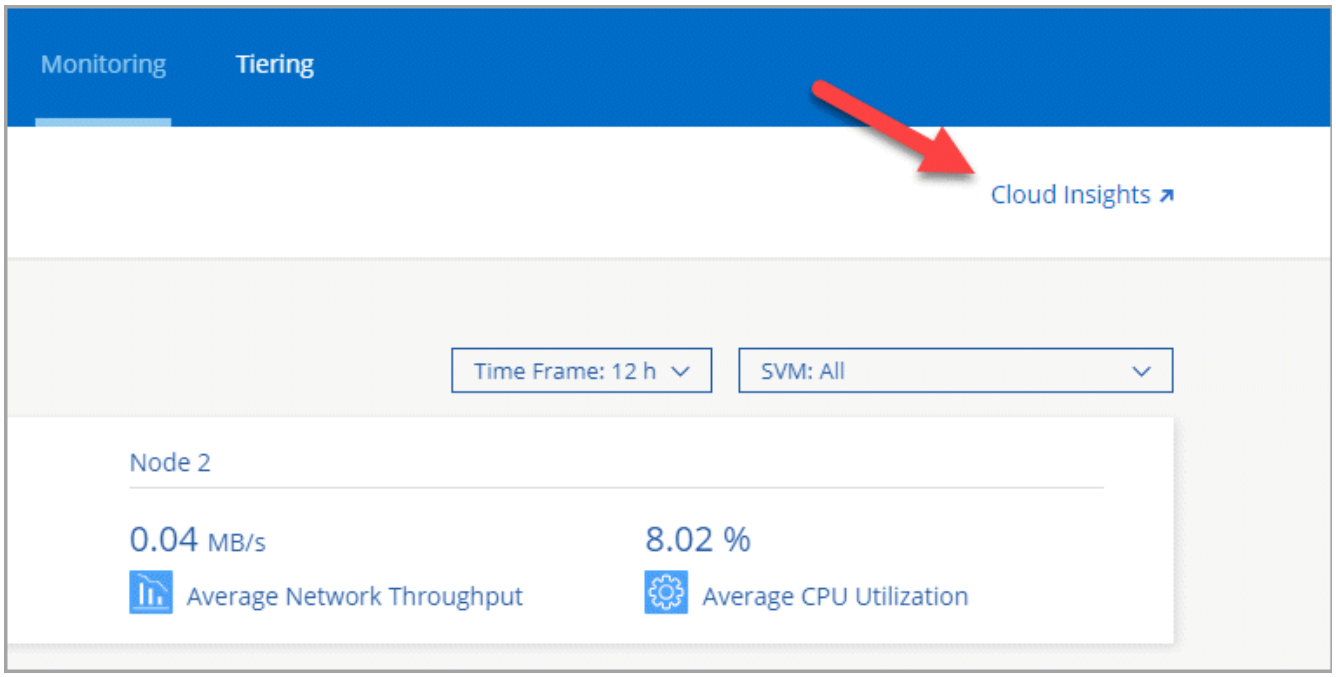
4. Use the data to identify performance issues to minimize impact on your users and apps.

Getting more information from Cloud Insights

The Monitoring tab in Cloud Manager provides basic performance data for your volumes. You can go to the Cloud Insights web interface from your browser to perform more in-depth monitoring and to configure alerts for your Cloud Volumes ONTAP systems.

Steps

1. At the top of Cloud Manager, click **Monitoring**.
2. Click the **Cloud Insights** link.



Result

Cloud Insights open in a new browser tab. If you need help, refer to the Cloud Insights documentation.

Disabling monitoring

If you no longer want to monitor Cloud Volumes ONTAP, you can disable the service at any time.

> If you disable monitoring from each of your working environments, you'll need to delete the EC2 instance yourself. The instance is named *AcquisitionUnit* with a generated hash (UUID) concatenated to it. For example: *AcquisitionUnit-FAN7FqeH*

Steps

1. At the top of Cloud Manager, click **Working Environments**.
2. Select a working environment.
3. In the pane on the right, click the ⋮ icon and select **Deactivate Scan**.

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.